

Physical Design Report

Innovus Flow - Academic Project

Design Block: `croc_chip`

Technology:	IHP SG13G2 (130nm)
Target Frequency:	100 MHz (10.0 ns period)
Clock Uncertainty:	0.1 ns
EDA Tool:	Cadence Innovus

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1. Overview and Setup

This report details the physical design (PD) flow for the `croc_chip` design block using the IHP SG13G2 technology node.

Timing constraints are defined in the `constraints.sdc` file, establishing a target frequency of 100 MHz ($T = 10.0$ ns), alongside an uncertainty of 0.1 ns for both setup and hold timing.

```
1 set TCK_SYS 10.0
2 create_clock -name clk_sys -period $TCK_SYS [get_ports clk_i]
3
4 set_clock_uncertainty 0.1 -setup [all_clocks]
5 set_clock_uncertainty 0.1 -hold [all_clocks]
6 set_clock_transition 0.2 [all_clocks]
```

Listing 1: Timing constraints

2. Floorplanning (00 & 01)

2.1. Core and Die Sizes

The Floorplan stage initializes the core and die dimensions, establishing areas for standard cell placement. A margin of $168\ \mu\text{m}$ from the core to the die edge is applied to all four sides to reserve space for I/O rings or pads.

```
1 floorPlan -site CoreSite -d 1840.32 1840.02 168 168 168 168
```

Listing 2: Floorplan initialization

- **Die Area:** $1840.32 \times 1840.02\ \mu\text{m}$
- **Core Area:** Calculated automatically based on the $168\ \mu\text{m}$ margins.
- **Target Utilization:** Approximately 58%.

2.2. Routing Layer Constraints

The design utilizes metal layers specified in the LEF file. The table below outlines the width, spacing, and pitch specifications for the 7 primary metal layers used for both signal routing and the Power/Ground (PG) mesh.

Table 1: Metal Layer constraints from LEF

Layer	Direction	Pitch (μm)	Width (μm)	Spacing (μm)
Metal1	Vertical	0.48	0.16	0.18
Metal2	Horizontal	0.42	0.20	0.22
Metal3	Vertical	0.48	0.20	0.22
Metal4	Horizontal	0.42	0.20	0.22
Metal5	Vertical	0.48	0.20	0.42
TopMetal1	Horizontal	2.28	1.64	1.64
TopMetal2	Vertical	4.00	2.00	2.00

2.3. Power and Ground (PG) Mesh

The Power/Ground (PG) Mesh is constructed on the upper metal layers to ensure stable power distribution. The PG Mesh is designed to route VDD and VSS through Metal3, Metal4, Metal5, TopMetal1, and TopMetal2. The table below specifies the width, set-to-set distance, and spacing for each metal layer:

Table 2: PG Mesh specifications

Layer	Direction	Width (μm)	Set-to-set (μm)	Spacing (μm)
Metal3	Vertical	1	15	2
Metal4	Horizontal	1	15	2
Metal5	Vertical	1	15	2
TopMetal1	Horizontal	4	30	4
TopMetal2	Vertical	4	30	4

The `addstripe` commands configure the thickness, direction, distances, and interconnecting vias between these layers.

```

1 # Metal3 (Vertical)
2 setAddStripeMode -reset
3 setAddStripeMode -stacked_via_top_layer Metal3
  -stacked_via_bottom_layer Metal1 -stapling_nets_style
  side_to_side
4 addstripe -layer Metal3 -direction vertical -nets {VDD VSS} -width 1
  -set_to_set_distance 15 -spacing 2 -start 0.5
5
6 # Metal4 (Horizontal)
7 setAddStripeMode -reset
8 setAddStripeMode -stacked_via_top_layer Metal4
  -stacked_via_bottom_layer Metal3 -stapling_nets_style
  side_to_side
9 addstripe -layer Metal4 -direction horizontal -nets {VDD VSS} -width
  1 -set_to_set_distance 15 -spacing 2 -start 0.5
10
11 # Metal5 (Vertical)
12 setAddStripeMode -stacked_via_top_layer Metal5
  -stacked_via_bottom_layer Metal4 -stapling_nets_style
  side_to_side
13 addstripe -layer Metal5 -direction vertical -nets {VDD VSS} -width
  1 -set_to_set_distance 15 -spacing 2 -start 0.5
14
15 # TopMetal1 (Horizontal)
16 setAddStripeMode -stacked_via_top_layer TopMetal1
  -stacked_via_bottom_layer Metal5 -stapling_nets_style
  side_to_side
17 addstripe -layer TopMetal1 -direction horizontal -nets {VDD VSS}
  -width 4 -set_to_set_distance 30 -spacing 4 -start 0.5
18
19 # TopMetal2 (Vertical)
20 setAddStripeMode -stacked_via_top_layer TopMetal2
  -stacked_via_bottom_layer TopMetal1 -stapling_nets_style

```

```
side_to_side
21 addstripe -layer TopMetal2 -direction vertical -nets {VDD VSS}
   -width 4 -set_to_set_distance 30 -spacing 4
```

Listing 3: PG Mesh setup



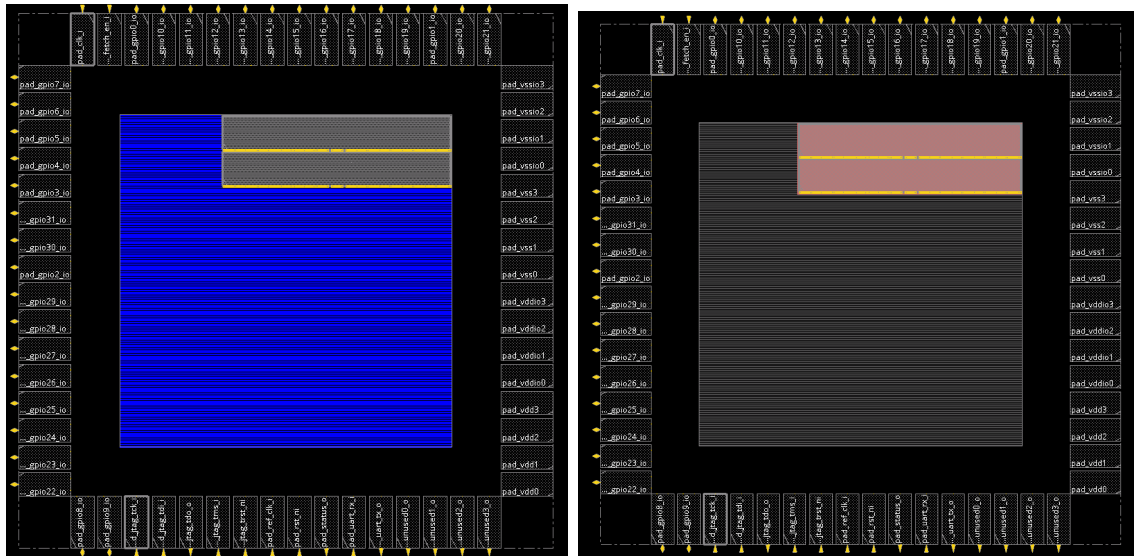


Figure 2: Metal1 Power Rails (left) and Full Floorplan (right)

3. Placement (02)

The Placement stage maps and positions all standard cells and hierarchical logic blocks into the designated rows. Logic blocks interacting with SRAMs are explicitly placed adjacent to their respective SRAM boundaries to optimize routing and minimize wirelength delays.

```

1 place_opt_design
2
3 setTieHiLoMode -reset
4 setTieHiLoMode -cell {sg13g2_tiehi sg13g2_tielo} -maxFanOut 10
5 addTieHiLo -cell {sg13g2_tiehi sg13g2_tielo} -prefix TIE

```

Listing 4: Placement execution and Tie Cells insertion

3.1. SRAM and Logic Block Placement

The figures below illustrate the logic blocks (highlighted colored cells) optimally placed near their corresponding SRAM blocks. The SRAM blocks are also encapsulated by placement blockages (Halos) to prevent DRC violations.

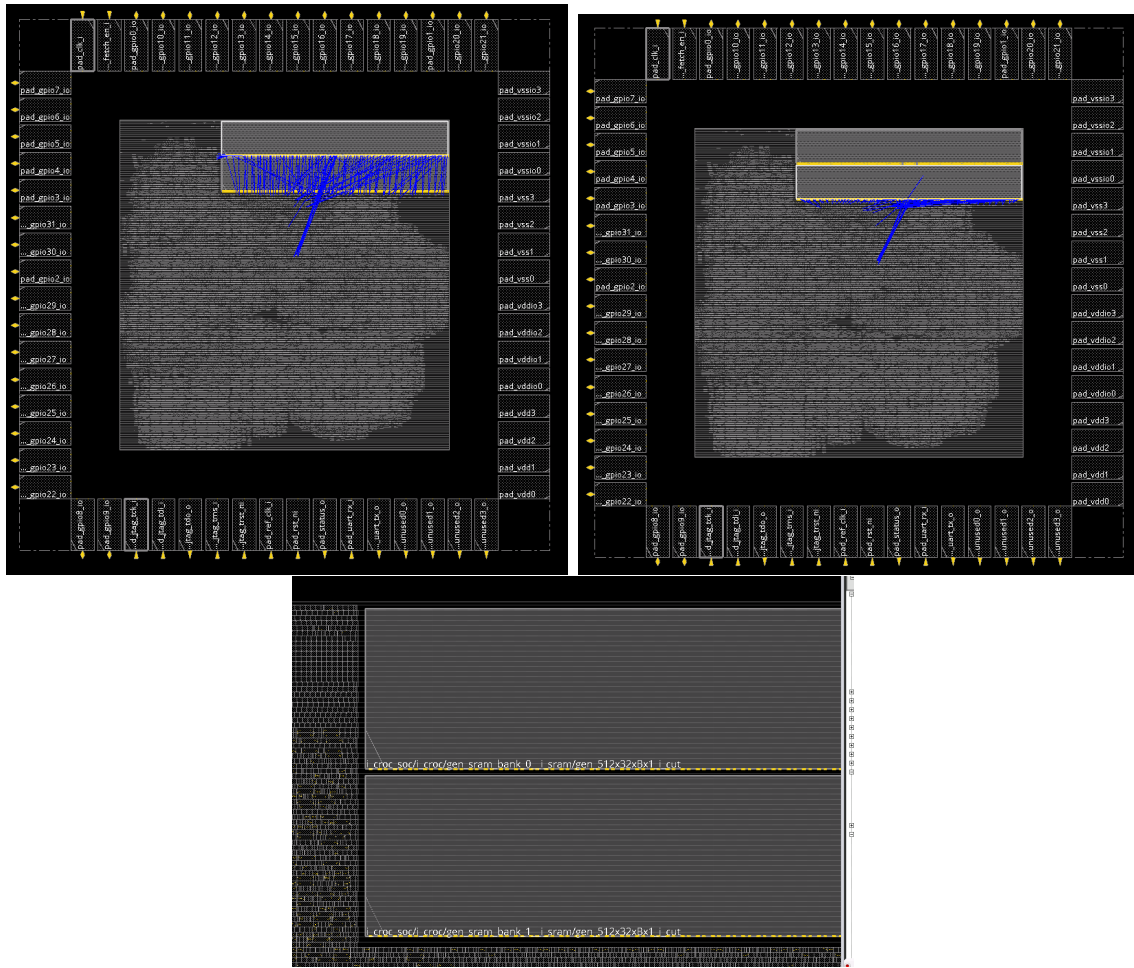


Figure 3: SRAM placement, adjacent logic cells, and Halo regions

3.2. Redundant Buffer Removal (ECO)

Post-placement timing reports indicated the presence of 9 redundant buffers that negatively impacted the critical paths. These timing violations were manually rectified using ECO (Engineering Change Order) commands by explicitly deleting the redundant instances in batch mode.

```

1 setEcoMode -batchMode true
2 ecoDeleteRepeater -inst <ten_instance_buffer_1>
3 ecoDeleteRepeater -inst <ten_instance_buffer_2>
4 # ... repeat to remove all 9 redundant buffers ...
5 setEcoMode -batchMode false

```

Listing 5: Redundant buffer removal (ECO script)

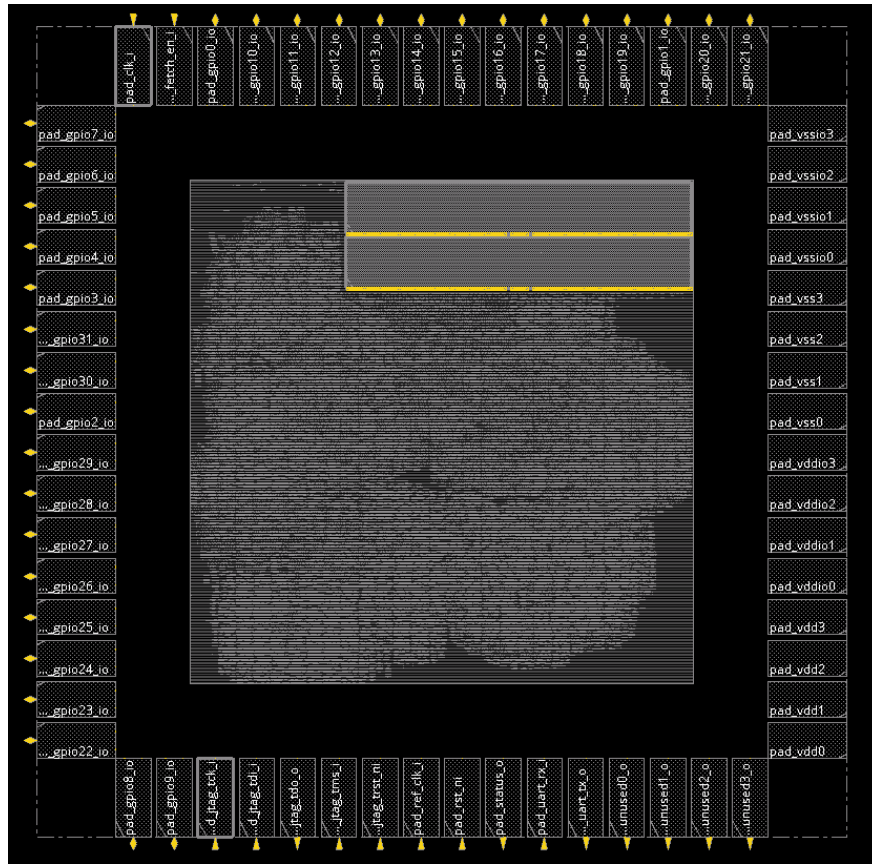


Figure 4: Full chip overview after Placement

3.3. Post-Placement Timing Report

The table below summarizes the timing results following placement (internal timing paths only). Paths involving I/O ports have been excluded. At this stage, the `reg2reg` path exhibits a slight negative slack (WNS = -0.033 ns).

Table 3: Internal Timing Summary (Post-Placement)

Path Type	WNS (ns)	TNS (ns)	Violating Paths (Vio)
reg → reg	-0.033	-0.366	33
reg → mem	0.719	0.000	0
mem → reg	0.054	0.000	0

4. Clock Tree Synthesis (CTS - 03 & 04)

CTS is the phase where the clock distribution network is synthesized to deliver the clock signal to all sequential elements (registers) with minimum skew and balanced latency. The `optDesign -postCTS` command performs further optimization, effectively driving the `reg2reg` timing slack to a positive value.

```

1 clockDesign -outDir ../rpt/${STAGE}/${STAGE}_clk
2 optDesign -postCTS
3 optDesign -postCTS -hold

```

Listing 6: CTS Execution and Optimization

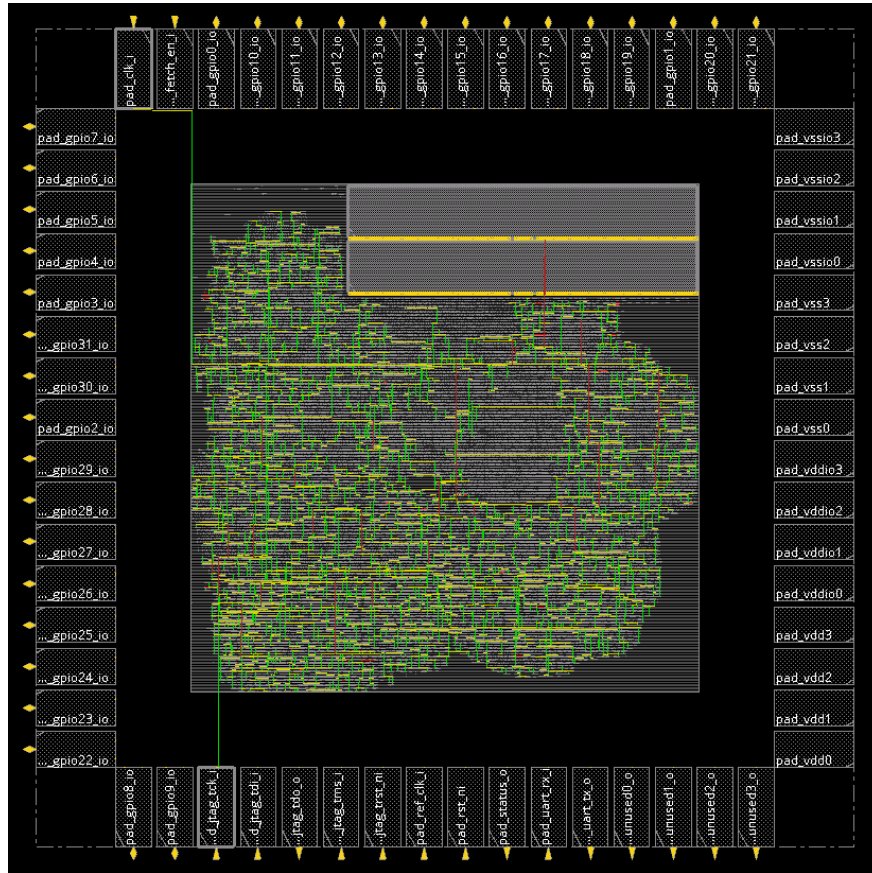


Figure 5: Clock Tree Routing visualization

4.1. Post-CTS Timing Report

The results from `optDesign` post-CTS demonstrate successful optimization, as there are no longer any violating paths in the `reg2reg` domain (WNS is positive). Any remaining violations (such as the -2.700 ns WNS seen in the overall summary) are strictly constrained to I/O communications.

Table 4: Internal Timing Summary (Post-CTS)

Path Type	WNS (ns)	TNS (ns)	Violating Paths (Vio)
reg → reg	0.001	0.000	0
reg → mem	0.638	0.000	0
mem → reg	0.044	0.000	0

5. Routing (05 & 06)

The Innovus NanoRoute engine executes both Global and Detailed Routing. Based on the `config.tcl` specifications, the available routing layers are strictly constrained from

Metal1 up to Metal5. This constraint fully dedicates the uppermost layers, TopMetal1 and TopMetal2, to the Power/Ground (PG) Mesh.

```
1 setDesignMode -topRoutingLayer Metal5
2 setDesignMode -bottomRoutingLayer Metal1
```

Listing 7: Routing layer constraints

Additionally, the NanoRoute configuration enables timing-driven routing and mitigates signal integrity issues (SI-driven).

```
1 setNanoRouteMode -quiet -routeWithTimingDriven 1
2 setNanoRouteMode -quiet -routeWithSiDriven 1
3 setNanoRouteMode -routeExpAdvancedPinAccess 2
4 routeDesign -globalDetail
```

Listing 8: NanoRoute configuration and execution

5.1. Useful Skew Optimization

To resolve the final timing violations during the post-route stage, **Useful Skew** optimization was enabled in the `optDesign -postRoute` command. Instead of strictly enforcing simultaneous clock arrivals, Useful Skew intentionally advances or delays the clock at specific endpoints, effectively "borrowing" time to compensate for data path delays.

```
1 setOptMode -usefulSkew true
2 setOptMode -usefulSkewPostRoute true
3 setOptMode -addInstancePrefix ictc_postRoute_extra_
4 optDesign -postRoute
5 optDesign -postRoute -hold
```

Listing 9: Useful Skew in Post-Route Optimization

5.2. Metal Layers and Finishing

The images below illustrate the individual routing metal layers and the insertion of Filler cells to bridge gaps between standard cells, thus preventing DRC well violations.

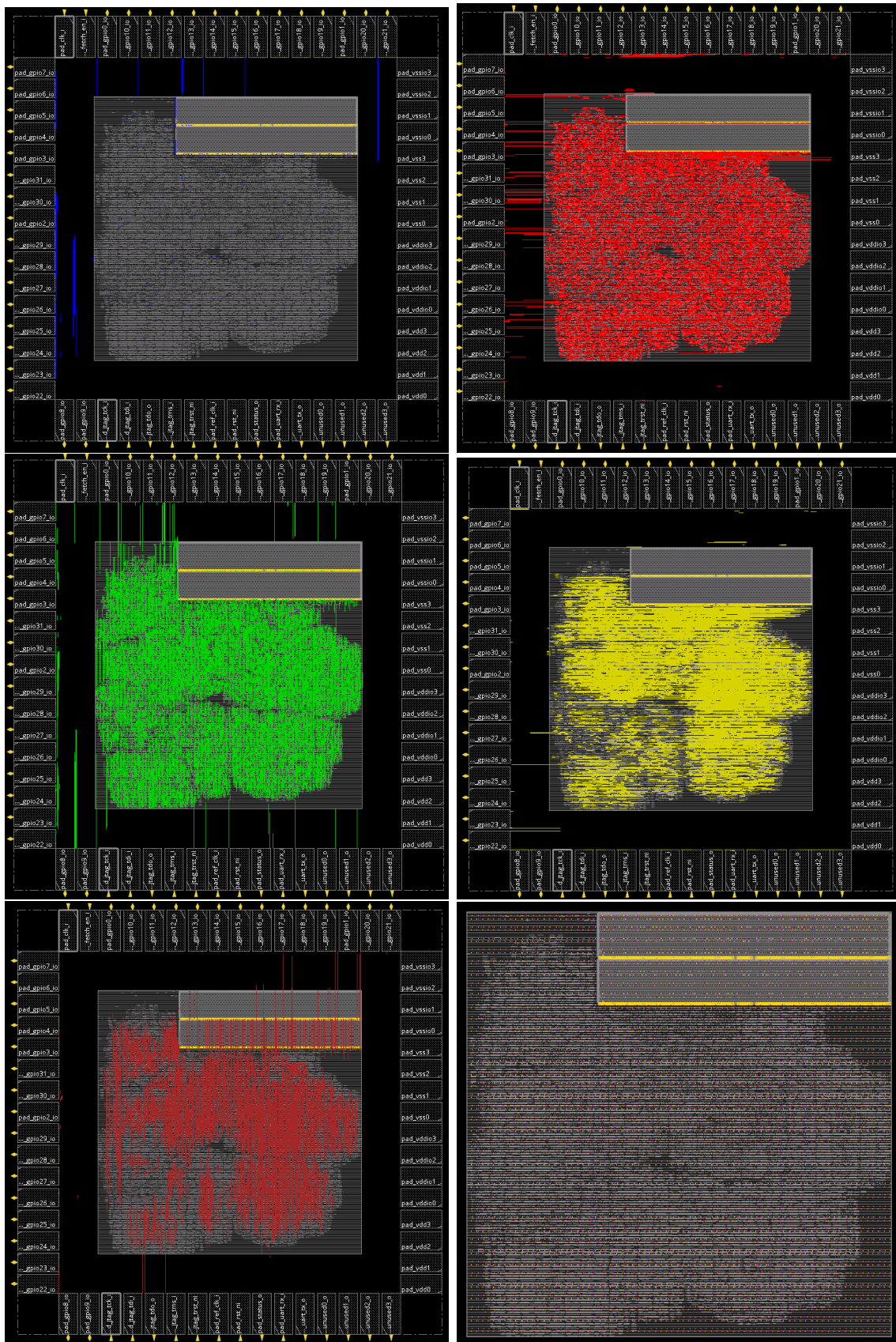


Figure 6: Routing progressively from M1 → M5 and Vias

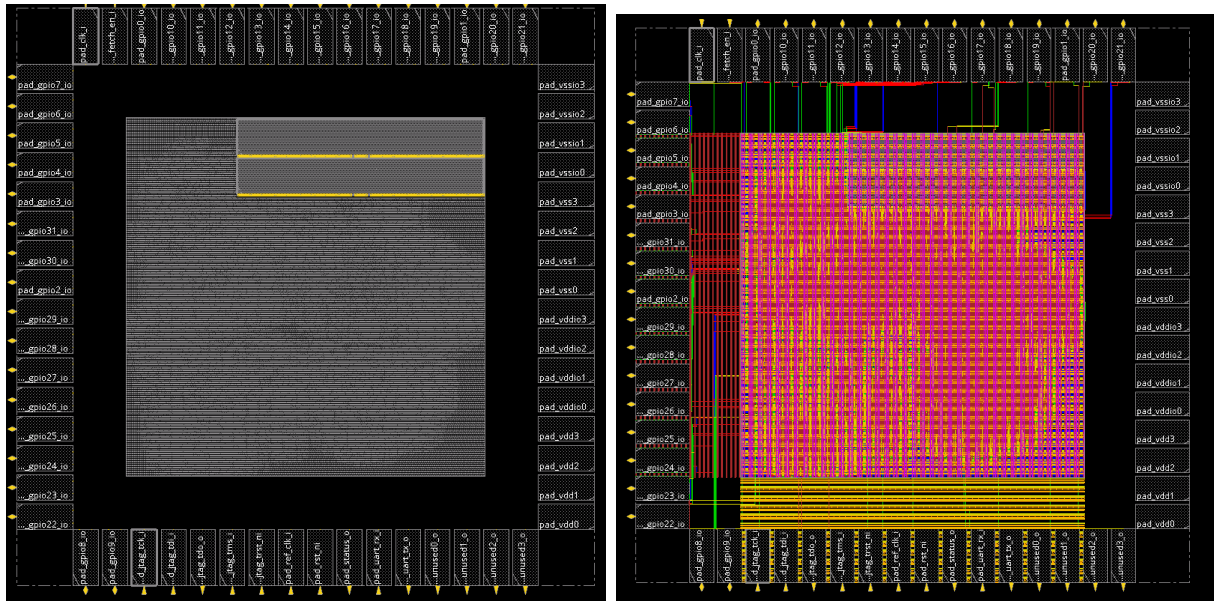


Figure 7: Filler Cells insertion (left) and final chip overview (right)